Railway Management Analytics

Data Analysis and Business Intelligence Project

Prepared by

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Data Analyst Portfolio Project

Technologies Used	Domain	Report Version
Python | Pandas | NumPy SQL | MySQL | Power BI Microsoft Excel | Git & GitHub	Transportation Analytics	Version 1.0 July 2026

Confidentiality Statement

This report has been prepared solely for educational and portfolio purposes. The datasets used are intended to demonstrate practical data analysis techniques, SQL querying, dashboard development, and business intelligence reporting. Any resemblance to operational railway data is purely for analytical demonstration.

Executive Summary

Introduction

Railway transportation is one of the most critical components of public infrastructure, serving millions of passengers every day. Efficient railway management depends on accurate data analysis to monitor operational performance, improve passenger satisfaction, optimize revenue, and reduce delays.

This project applies data analytics techniques to analyze railway operational data and transform raw information into meaningful business insights. The project demonstrates the complete analytics lifecycle—from data collection and cleaning to SQL analysis, interactive dashboard creation, and business recommendations.

The analysis combines Python for data preprocessing, SQL for business querying, and Power BI for interactive visualization, providing a comprehensive business intelligence solution for railway operations.

Project Objectives

The primary objective of this project is to evaluate railway operational performance by analyzing passenger bookings, train schedules, revenue generation, delays, route efficiency, and station activity.

The project aims to answer several business-critical questions, including:

- Which routes generate the highest revenue?
- Which stations experience the highest passenger traffic?
- How do booking patterns change over time?
- Which trains are most frequently delayed?
- Which travel classes contribute the most revenue?
- What seasonal trends influence railway operations?

Dataset Summary

The project integrates multiple datasets representing different operational components of a railway management system.

Dataset	Purpose
Passengers	Passenger demographic information
Bookings	Ticket booking transactions
Tickets	Fare and travel details
Trains	Train information
Routes	Route mapping
Stations	Station details
Train Schedule	Arrival and departure schedules
Delays	Delay records
Payments	Payment transactions
Employees	Staff information
Maintenance	Maintenance activities

Tools and Technologies

Tool	Purpose

Python	Data Cleaning
Pandas	Data Manipulation
NumPy	Numerical Operations
SQL	Business Queries
Excel	Initial Data Validation
Power BI	Dashboard Development
GitHub	Version Control

Key Performance Indicators (KPIs)

The dashboard was designed around business-focused KPIs, including:

- Total Revenue
- Total Bookings
- Total Passengers
- Average Ticket Fare
- Route Performance
- Delay Percentage
- Passenger Distribution
- Monthly Revenue Trend
- Booking Trend
- Station Performance

Business Impact

The analytical insights generated from this project can support railway management in:

- Improving train punctuality
- Optimizing resource allocation
- Increasing operational efficiency
- Enhancing passenger satisfaction
- Identifying profitable routes
- Improving scheduling decisions
- Supporting data-driven planning

Table of Contents

1. Executive Summary
2. Business Understanding
3. Data Understanding
4. Data Cleaning and Preparation
5. Exploratory Data Analysis (EDA)
6. SQL Business Analysis
7. Power BI Dashboard Development
8. KPI Analysis and Business Insights
9. Business Recommendations and Project Conclusion
10. Appendix

2. Business Problem

2.1 Background

Railway transportation is one of the most widely used modes of public transport, serving millions of passengers every day. Efficient railway operations depend on accurate scheduling, optimal resource utilization, punctual train services, and effective revenue management. As railway networks continue to expand, managing large volumes of operational data becomes increasingly challenging.

Traditional reporting methods often provide only historical summaries, making it difficult for management to quickly identify operational issues or emerging trends. Modern railway organizations require data-driven decision-making supported by business intelligence solutions that consolidate information from multiple operational systems.

2.2 Business Problem Statement

Railway management faces several operational and strategic challenges due to fragmented data and limited visibility into key performance indicators. Without integrated analytics, management struggles to answer critical business questions such as:

- Which routes generate the highest revenue?
- Which stations experience the highest passenger traffic?
- Which trains have recurring delays?
- How do booking trends vary across months and travel classes?
- Which factors contribute most to operational inefficiencies?

2.3 Problem Identification

Operational Challenges

- Frequent train delays affecting service reliability.
- Difficulty identifying consistently underperforming routes.
- Limited visibility into station-level performance.
- Inconsistent monitoring of train schedules.

Revenue Challenges

- Uneven revenue generation across routes and travel classes.
- Limited understanding of seasonal booking patterns.
- Difficulty identifying high-value customer segments.

Customer Experience Challenges

- Delays leading to reduced passenger satisfaction.
- Overcrowding during peak travel periods.
- Inadequate planning for high-demand routes.

Data Management Challenges

- Data stored across multiple datasets.
- Inconsistent data formats.
- Missing and duplicate records.
- Manual reporting processes prone to errors.

3. Business Objectives

The primary objective is to transform raw railway operational data into meaningful business intelligence that supports informed decision-making.

Objective 1: Analyze Passenger Demand

Understand passenger travel behavior by examining booking volumes, travel classes, and station activity.

Expected Outcome: Identify high-demand routes and peak travel periods.

Objective 2: Evaluate Revenue Performance

Measure revenue generated across different routes, stations, and travel classes.

Expected Outcome: Identify the most profitable services and opportunities for revenue optimization.

Objective 3: Monitor Train Punctuality

Analyze delay records to identify recurring operational issues.

Expected Outcome: Highlight trains and routes requiring operational improvements.

Objective 4: Assess Route Performance

Evaluate route utilization using passenger volume, booking frequency, and revenue.

Expected Outcome: Support decisions related to scheduling and resource allocation.

Objective 5: Improve Station Operations

Analyze passenger traffic and booking activity across stations.

Expected Outcome: Identify stations requiring additional infrastructure or staffing.

Objective 6: Develop Interactive Dashboards

Design Power BI dashboards that enable users to explore KPIs through filters and visualizations.

Expected Outcome: Provide stakeholders with an intuitive tool for monitoring railway performance.

4. Project Scope

4.1 In Scope

- Data collection and integration.
- Data cleaning and preprocessing using Python.

- SQL-based business analysis.
- Exploratory Data Analysis (EDA).
- KPI calculation.
- Power BI dashboard development.
- Business insight generation.
- Recommendation of operational improvements.

4.2 Out of Scope

- Real-time train tracking.
- Live passenger booking systems.
- Predictive maintenance using IoT data.
- Machine learning-based demand forecasting.
- Mobile application development.
- Integration with external APIs.

5. Stakeholder Analysis

Stakeholder	Role	Information Required
Railway Management	Strategic planning	Revenue, KPIs, profitability
Operations Team	Daily operations	Delays, schedules, train performance
Finance Department	Financial analysis	Revenue, ticket sales, payment trends
Station Managers	Station management	Passenger traffic, booking activity
Route Planners	Route optimization	Demand, occupancy, route performance
Customer Service Team	Passenger support	Delay information, booking trends

6. Business Questions

Revenue Analysis

- Which routes generate the highest revenue?
- Which travel class contributes the most to total revenue?
- How does monthly revenue change over time?

Passenger Analysis

- Which stations have the highest passenger traffic?
- Which cities contribute the largest number of passengers?
- Which months experience the highest booking volume?

Delay Analysis

- Which trains are delayed most frequently?
- Which routes experience the highest average delays?
- Are delays concentrated during specific periods?

Booking Analysis

- Which booking methods are most frequently used?
- What are the monthly booking trends?

- Which travel classes are most popular?

Route Performance

- Which routes are most frequently traveled?
- Which routes generate high revenue but low passenger volume?
- Which routes require operational improvements?

7. Project Deliverables

Cleaned Dataset

A validated and integrated dataset prepared for analysis.

SQL Scripts

Queries used for business analysis and KPI generation.

Power BI Dashboard

Interactive dashboards featuring KPIs, trends, filters, and drill-through capabilities.

Analytical Report

A comprehensive report documenting the project methodology, findings, and recommendations.

GitHub Repository

A well-structured repository containing datasets, notebooks, SQL scripts, dashboard files, documentation, and the project README.

8. Success Metrics

Metric	Description
Data Quality	Missing values and duplicates resolved
Dashboard Accuracy	KPIs match source data
Query Performance	SQL queries return correct results efficiently
Business Insights	Findings support operational decisions
Dashboard Usability	Easy navigation and interactive filtering
Documentation	Complete and reproducible project documentation

3. Data Understanding

3.1 Introduction

Data understanding is a critical phase in the analytics lifecycle. Before performing any analysis, it is essential to understand the structure, quality, and relationships within the data. This phase involved reviewing the available datasets, examining their attributes, identifying potential data quality issues, and understanding how the datasets relate to one another.

3.2 Data Source

Attribute	Description
Domain	Railway Transportation
Data Type	Structured Data
Format	CSV Files
Number of Files	11
Storage	Local File System
Analysis Tools	Python, SQL, Power BI
Database Compatibility	MySQL

3.3 Dataset Overview

Dataset	Description	Business Purpose
Passengers	Passenger demographic information	Customer Analysis
Bookings	Booking transactions	Booking Trend Analysis
Tickets	Ticket and fare information	Revenue Analysis
Trains	Train master information	Train Performance
Train Schedule	Arrival and departure schedules	Operational Analysis
Routes	Route details	Route Performance
Stations	Station information	Station Analysis
Payments	Payment transactions	Financial Analysis
Delays	Delay records	Delay Analysis
Employees	Employee information	Resource Monitoring
Maintenance	Maintenance records	Asset Management

3.4 Dataset Statistics

Dataset	Approximate Records
Passengers	5,000
Bookings	8,000
Tickets	8,000
Trains	200
Routes	150
Stations	75
Train Schedule	1,500
Payments	8,000
Delays	1,500
Employees	500

Maintenance	400

Total Records: More than 30,000 records across all datasets.

3.5 Dataset Relationships

Parent Table	Child Table	Relationship
Passengers	Bookings	One-to-Many
Bookings	Tickets	One-to-One / One-to-Many
Trains	Train Schedule	One-to-Many
Routes	Trains	One-to-Many
Stations	Routes	One-to-Many
Bookings	Payments	One-to-One
Trains	Delays	One-to-Many
Trains	Maintenance	One-to-Many

3.6 Data Dictionary

3.6.1 Passengers Dataset

Column	Data Type	Description
passenger_id	Integer	Unique passenger identifier
first_name	Text	Passenger first name
last_name	Text	Passenger last name
gender	Text	Gender
age	Integer	Passenger age
city	Text	City of residence
state	Text	State
phone	Text	Contact number
email	Text	Email address
registration_date	Date	Registration date

3.6.2 Bookings Dataset

Column	Data Type	Description
booking_id	Integer	Booking identifier
passenger_id	Integer	Passenger reference
train_id	Integer	Train reference
booking_date	Date	Booking date
journey_date	Date	Travel date
class	Text	Travel class
booking_status	Text	Booking status
total_amount	Decimal	Booking amount
payment_id	Integer	Payment reference

3.6.3 Tickets Dataset

Column	Data Type	Description
ticket_id	Integer	Ticket ID
booking_id	Integer	Booking reference
seat_number	Text	Seat number
coach	Text	Coach
fare	Decimal	Ticket fare
ticket_status	Text	Ticket status
travel_class	Text	Travel class

3.6.4 Trains Dataset

Column	Description
train_id	Unique train identifier
train_name	Train name
route_id	Route reference
train_type	Express / Superfast / Passenger
total_seats	Available seating capacity
status	Operational status

3.6.5 Routes Dataset

Column	Description
route_id	Route identifier
source_station	Source station
destination_station	Destination station
distance_km	Distance in kilometers
estimated_time	Journey duration

3.6.6 Stations Dataset

Column	Description
station_id	Station identifier
station_name	Station name
city	City
state	State
zone	Railway zone

3.6.7 Payments Dataset

Column	Description
payment_id	Payment identifier
booking_id	Booking reference
payment_mode	UPI/Card/Cash/Net Banking
payment_date	Payment date
amount	Paid amount

3.6.8 Train Schedule Dataset

Column	Description
schedule_id	Schedule identifier
train_id	Train reference
arrival_time	Arrival time
departure_time	Departure time
platform	Platform number

3.6.9 Delays Dataset

Column	Description
delay_id	Delay identifier
train_id	Train reference
delay_minutes	Delay duration
delay_reason	Reason for delay
delay_date	Delay occurrence date

3.6.10 Employees Dataset

Column	Description
employee_id	Employee identifier
employee_name	Employee name
department	Department
designation	Job title
station_id	Assigned station

3.6.11 Maintenance Dataset

Column	Description
maintenance_id	Maintenance identifier
train_id	Train reference
maintenance_type	Maintenance category
maintenance_date	Service date
maintenance_cost	Maintenance expense

3.7 Entity Relationship (ER) Diagram

```
Passengers
     |
     +--> Bookings
               |
               +--> Tickets
               +--> Payments

Trains
   |
   +--> Train Schedule
   +--> Delays
   +--> Maintenance

Routes --> Trains
Stations --> Routes
Employees --> Stations
```

Figure 3.1: Entity Relationship Diagram of the Railway Management Analytics Database.

3.8 Data Flow Architecture

```
CSV Files --> Python (Pandas) --> Data Cleaning --> Feature Engineering
         --> Data Validation --> MySQL Database --> SQL Analysis
         --> Power BI --> Interactive Dashboard --> Business Insights
```

Figure 3.2: Data Flow Architecture of the Railway Management Analytics Project.

3.9 Data Quality Assessment

Identified Issues

- Missing values in selected numerical and categorical columns.
- Duplicate records affecting transaction counts.
- Inconsistent date formats across datasets.
- Variations in text formatting (uppercase/lowercase, extra spaces).
- Potential orphan records caused by missing foreign key references.

Data Quality Improvements

- Missing values were imputed using appropriate methods.

- Duplicate records were removed.
- Date fields were standardized to a consistent format.
- Text values were normalized.
- Column names were standardized.
- Relationships were validated before merging datasets.

4. Data Cleaning and Preparation

4.1 Introduction

Raw operational data often contains inconsistencies such as missing values, duplicate records, incorrect data types, inconsistent formatting, and invalid relationships. The objective of this phase was to transform the raw railway datasets into a clean, consistent, and analysis-ready format using Python and the Pandas library.

4.3 Importing Required Libraries

```
import pandas as pd
import numpy as np
```

4.4 Loading the Datasets

```
passengers = pd.read_csv("passengers.csv")
bookings = pd.read_csv("bookings.csv")
tickets = pd.read_csv("tickets.csv")
trains = pd.read_csv("trains.csv")
stations = pd.read_csv("stations.csv")
routes = pd.read_csv("routes.csv")
payments = pd.read_csv("payments.csv")
train_schedule = pd.read_csv("train_schedule.csv")
delays = pd.read_csv("delays.csv")
employees = pd.read_csv("employees.csv")
maintenance = pd.read_csv("maintenance.csv")
```

4.5 Initial Data Inspection

```
bookings.info()
bookings.head()
bookings.describe()
```

4.6 Handling Missing Values

```
# Identify
bookings.isnull().sum()

# Fill Numerical
bookings["total_amount"] =
bookings["total_amount"].fillna(bookings["total_amount"].mean())
tickets["fare"] = tickets["fare"].fillna(tickets["fare"].mean())

# Fill Categorical
passengers["city"] = passengers["city"].fillna("Unknown")
```

4.7 Removing Duplicate Records

```
bookings.duplicated().sum()
bookings.drop_duplicates(inplace=True)
```

4.8 Standardizing Column Names

```
bookings.columns = bookings.columns.str.lower().str.strip().str.replace(" ", "_")
```

4.9 Data Type Conversion

```
bookings["booking_date"] = pd.to_datetime(bookings["booking_date"])
payments["payment_date"] = pd.to_datetime(payments["payment_date"])
```

4.10 Text Standardization

```
passengers["city"] = passengers["city"].str.strip().str.title()
```

4.11 Feature Engineering

```
bookings["booking_month"] = bookings["booking_date"].dt.month_name()
bookings["booking_year"] = bookings["booking_date"].dt.year
bookings["day_name"] = bookings["booking_date"].dt.day_name()
bookings["weekend_booking"] = np.where(
    bookings["booking_date"].dt.weekday >= 5, "Weekend", "Weekday"
)
```

4.12 Dataset Integration

```
passenger_booking = passengers.merge(bookings, on="passenger_id", how="inner")
passenger_booking_ticket = passenger_booking.merge(tickets, on="booking_id",
how="inner")
full_data = passenger_booking_ticket.merge(train_schedule, on="train_id",
how="left")
```

4.14 Exporting the Final Dataset

```
full_data.to_csv("railway_management_system_final.csv", index=False)
```

4.15 Data Quality Improvements

Data Quality Issue	Solution Implemented	Business Benefit
Missing values	Mean imputation / 'Unknown' category	Preserved records and reduced bias
Duplicate records	Removed duplicates	Accurate KPIs and reporting
Inconsistent column names	Standardized naming	Easier coding and SQL queries
Incorrect data types	Converted to appropriate formats	Enabled time-based analysis
Inconsistent text values	Standardized capitalization and spacing	Reliable grouping and filtering
Disconnected datasets	Merged using keys	Comprehensive cross-domain analysis

Section 4 Exploratory Data Analysis (EDA)

5. Exploratory Data Analysis (EDA)

5.1 Introduction

Exploratory Data Analysis (EDA) is a critical stage in the analytics process that involves examining the dataset to understand its characteristics, identify patterns, detect anomalies, and generate meaningful business insights. EDA was conducted to analyze passenger behavior, booking trends, revenue generation, route performance, station activity, train delays, and payment preferences.

5.4 Passenger Demographic Analysis

Age Distribution

Visualization: Histogram or Column Chart

Observation: Majority of passengers fall within the working-age group (21–40 years). Fewer bookings are made by children and senior citizens.

Business Insight: Working professionals represent the largest customer segment, suggesting that scheduling and service offerings should prioritize commuter needs.

Gender Distribution

Visualization: Pie Chart or Donut Chart

Observation: Passenger representation is relatively balanced with minor variations depending on travel routes.

Business Insight: No major gender imbalance indicates that railway services are utilized broadly across demographics.

City-wise Passenger Distribution

Visualization: Bar Chart

Observation: Major metropolitan areas contribute significantly more passengers. Smaller cities have lower booking volumes.

Business Insight: High-demand cities require greater train frequency and station capacity.

5.5 Booking Trend Analysis

Monthly Booking Trend

Visualization: Line Chart

Observation: Booking volumes fluctuate seasonally. Higher bookings are observed during holidays and festive periods.

Business Insight: Railway management can allocate additional coaches and trains during peak months.

Weekday vs Weekend Bookings

Visualization: Clustered Bar Chart

Observation: Weekend bookings are generally higher due to leisure travel. Weekday bookings reflect commuter demand.

Business Insight: Weekend services may require increased capacity to accommodate higher passenger volumes.

5.6 Revenue Analysis

5.7 Route & Station Performance Analysis

5.8 Train Delay Analysis

Delay Reasons

Common causes of delays may include:

- Technical issues.
- Maintenance activities.
- Weather conditions.
- Operational congestion.

5.9 Key Insights from EDA

- Passenger demand is concentrated in metropolitan regions and major railway junctions.
- Booking and revenue exhibit clear seasonal patterns, with peaks during holidays and festivals.
- Premium travel classes generate higher revenue despite lower passenger volumes.
- A small number of routes contribute a significant share of total revenue.
- Recurring delays are concentrated on specific trains and routes.
- Digital payment methods are widely adopted.
- Passenger traffic is unevenly distributed across stations.

6. SQL Business Analysis

6.1 Introduction

SQL was used to transform raw railway data into actionable business insights by querying, aggregating, and joining multiple datasets. The database was implemented in MySQL, and the cleaned dataset was imported after preprocessing in Python.

6.6 Basic SQL Queries

Query 1: View Passenger Data

```
SELECT *
FROM passengers;
```

Query 2: Retrieve Booking Information

```
SELECT booking_id,
       booking_date,
       total_amount
FROM bookings;
```

Query 3: Filter Confirmed Bookings

```
SELECT *
FROM bookings
WHERE booking_status = 'Confirmed';
```

Query 4: Sort Bookings by Revenue

```
SELECT booking_id,
       total_amount
FROM bookings
ORDER BY total_amount DESC;
```

6.7 Aggregate Functions

Total Revenue

```
SELECT SUM(total_amount) AS total_revenue
FROM bookings;
```

Average Ticket Fare

```
SELECT AVG(fare) AS average_fare
FROM tickets;
```

Total Passengers

```
SELECT COUNT(passenger_id) AS total_passengers
FROM passengers;
```

6.8 GROUP BY Analysis

Monthly Revenue

```sql
SELECT MONTH(booking_date) AS month,
       SUM(total_amount) AS revenue
FROM bookings
GROUP BY MONTH(booking_date)
ORDER BY month;
```

Revenue by Travel Class

```sql
SELECT class,
       SUM(total_amount) AS revenue
FROM bookings
GROUP BY class;
```

Passenger Count by City

```sql
SELECT city,
       COUNT(*) AS passengers
FROM passengers
GROUP BY city
ORDER BY passengers DESC;
```

6.9 JOIN Operations

Passenger Booking Details

```sql
SELECT p.passenger_id, p.first_name,
       b.booking_id, b.booking_date
FROM passengers p
INNER JOIN bookings b ON p.passenger_id = b.passenger_id;
```

Train Route Details

```sql
SELECT t.train_name,
       r.source_station, r.destination_station
FROM trains t
INNER JOIN routes r ON t.route_id = r.route_id;
```

6.11 CASE Statement

```sql
SELECT ticket_id, fare,
CASE
    WHEN fare < 500 THEN 'Low Fare'
    WHEN fare BETWEEN 500 AND 1000 THEN 'Medium Fare'
    ELSE 'High Fare'
END AS fare_category
FROM tickets;
```

6.13 Common Table Expression (CTE)

```sql
WITH MonthlyRevenue AS (
    SELECT MONTH(booking_date) AS month,
           SUM(total_amount) AS revenue
    FROM bookings
    GROUP BY MONTH(booking_date)
)
SELECT * FROM MonthlyRevenue;
```

6.14 Window Functions (MySQL 8.0+)

```sql
SELECT route_id,
       SUM(total_amount) AS revenue,
       RANK() OVER(ORDER BY SUM(total_amount) DESC) AS revenue_rank
```

```
FROM bookings
GROUP BY route_id;
```

6.15 Business Questions Answered Using SQL

Business Question	SQL Technique Used
What is total revenue?	SUM()
How many passengers are registered?	COUNT()
What is the average fare?	AVG()
Which city has the highest passengers?	GROUP BY
Which class generates the most revenue?	GROUP BY + SUM()
Which bookings are confirmed?	WHERE
Which routes perform best?	JOIN + GROUP BY
Which fares are above average?	Subquery
Which cities have more than 100 passengers?	HAVING
Which routes rank highest?	Window Function

Section 6 Power BI Dashboard Development

7. Power BI Dashboard Development

7.1 Introduction

Microsoft Power BI was used to create an interactive Railway Management Analytics Dashboard. The dashboard integrates cleaned railway data, calculated KPIs, and analytical insights generated from Python and SQL.

The dashboard provides railway stakeholders with a centralized platform to monitor revenue performance, passenger demand, booking trends, train operations, route efficiency, station performance, and delay patterns.

7.3 Power BI Development Workflow

```
Cleaned Dataset --> Data Import into Power BI --> Data Transformation
--> Data Modeling --> Relationship Creation --> DAX Measures
--> Visual Development --> Dashboard Testing --> Final Dashboard
```

7.4 Data Import into Power BI

The final cleaned dataset (railway_management_system_final.csv) was imported into Power BI for visualization and reporting.

Import Steps:

1. Open Power BI Desktop.
2. Select: Home > Get Data > Text/CSV
3. Select the cleaned railway dataset.
4. Load the data.
5. Verify column data types.

7.6 Data Modeling

7.6.2 Star Schema Design

The project follows a Star Schema approach with Fact_Booking as the central fact table.

Table Type	Table Name	Key Fields
Fact Table	Fact_Booking	Booking ID, Passenger ID, Train ID, Fare, Amount, Booking Date
Dimension	Dim_Passenger	Passenger ID, Age, Gender, City
Dimension	Dim_Train	Train ID, Train Name, Train Type
Dimension	Dim_Route	Route ID, Source, Destination, Distance
Dimension	Dim_Station	Station ID, Station Name, Location
Dimension	Dim_Date	Date, Month, Year, Quarter

7.8 DAX Measures Development

Total Revenue

```
Total Revenue =
SUM(Bookings[total_amount])
```

Total Tickets

```
Total Tickets =
COUNT(Tickets[ticket_id])
```

Total Passengers

```
Total Passengers =
DISTINCTCOUNT(Passengers[passenger_id])
```

Average Fare

```
Average Fare =
AVERAGE(Tickets[fare])
```

Avg Revenue Per Ticket

```
Avg Revenue Per Ticket =
DIVIDE([Total Revenue], [Total Tickets])
```

Delay Percentage

```
Delay Percentage =
DIVIDE(
    COUNT(Delays[delay_id]),
    COUNT(Train_Schedule[schedule_id])
)
```

Cancellation Rate

```
Cancellation Rate =
DIVIDE(
    COUNTROWS(FILTER(Bookings, Bookings[booking_status]="Cancelled")),
    COUNT(Bookings[booking_id])
)
```

7.9 Dashboard Structure – Overview

Page 1: Home Dashboard

- KPI Cards (Total Revenue, Total Tickets, Total Passengers, Average Fare)
- Revenue Trend – Line Chart
- Booking Trend – Column Chart
- Revenue by Class – Bar Chart
- Top Cities – Map / Bar Chart
- Filters: Year, Travel Class

Page 2: Passenger Analysis Dashboard

- Passenger Age Distribution
- Gender Distribution (Pie Chart)
- City-wise Passenger Count (Bar Chart)
- Travel Class Preference

Page 3: Revenue Analysis Dashboard

- Monthly Revenue – Line Chart
- Revenue by Route – Bar Chart

- Revenue by Class – Column Chart
- Payment Analysis

Page 4: Train Performance Dashboard

- Delay Count – Column Chart
- Average Delay Time – KPI Card
- Train-wise Performance – Bar Chart
- Route Delay Comparison

Page 5: Station and Route Dashboard

- Top Stations by Passenger Traffic – Bar Chart
- Route Popularity – Horizontal Bar Chart
- Distance vs Revenue Scatter Plot

7.10 Dashboard Screenshots

The following screenshots illustrate the interactive Power BI dashboards developed for the Railway Management Analytics project.

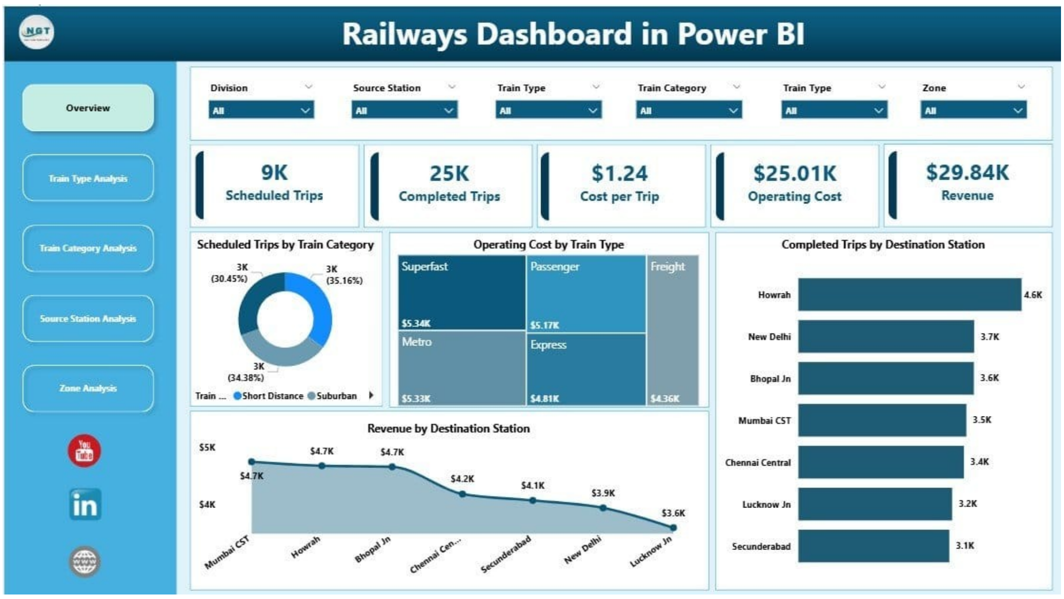

Figure 7.1: Railway Management Executive Performance Dashboard – Overview of KPIs, Revenue Trends, and Booking Analysis

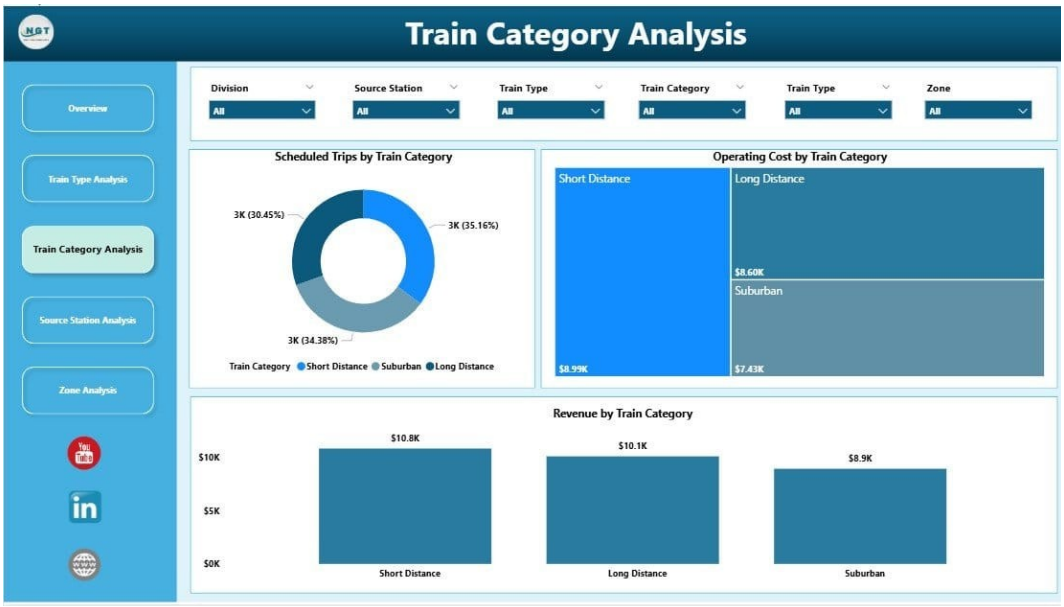

Figure 7.2: Passenger Analysis Dashboard – Booking Trends, Route Performance, and Delay Insights

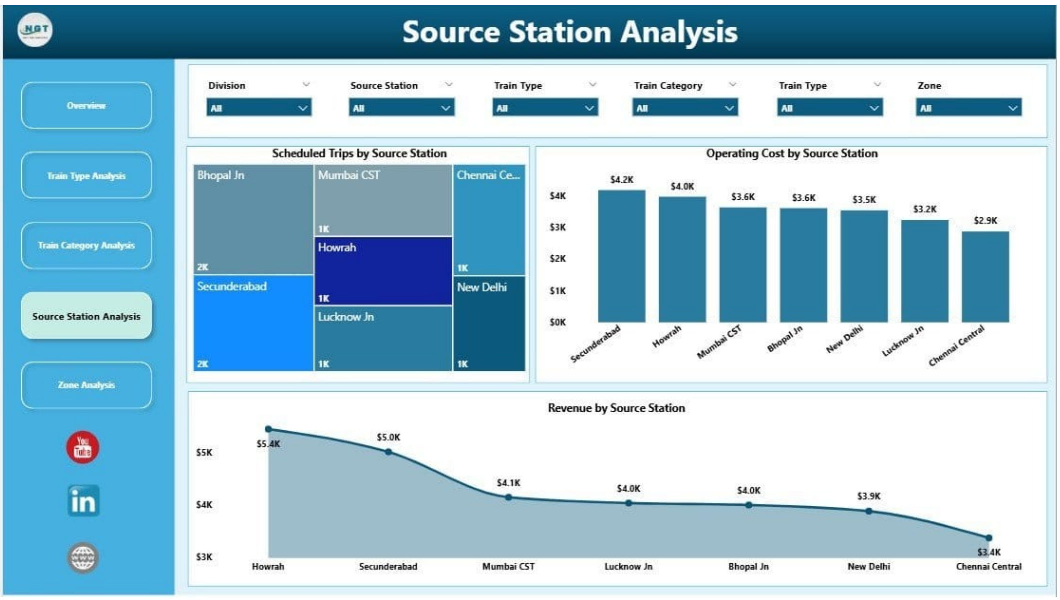

Figure 7.3: Revenue and Route Performance Dashboard – Revenue by Route and Travel Class

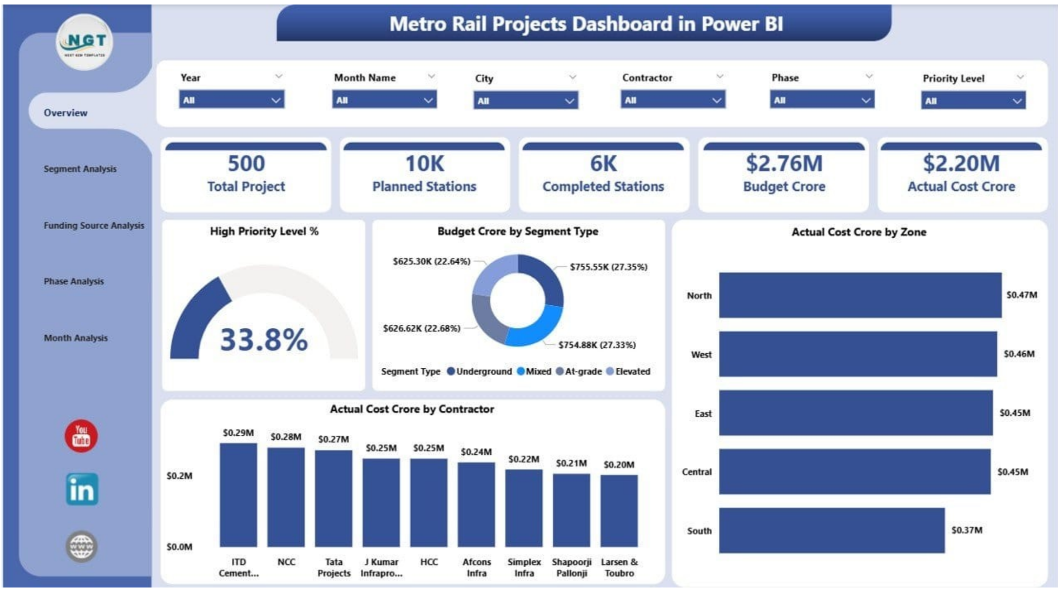

Figure 7.4: Metro Rail Projects Dashboard – Station and Route-Level Performance Monitoring

7.11 Dashboard Design Principles

Principle	Implementation
Consistency	Same fonts, similar spacing, standardized layouts
Simplicity	Avoided unnecessary visuals, focused on important KPIs
Interactivity	Slicers, Filters, Drill-through, Navigation buttons
Business Focus	Every visual answers a specific business question

7.12 Visualization Selection Strategy

Requirement	Visual Used
KPI monitoring	Cards
Trend analysis	Line Chart
Category comparison	Bar Chart
Composition analysis	Donut Chart
Geographic analysis	Map
Detailed comparison	Table

8. KPI Analysis and Business Insights

8.1 Introduction

Key Performance Indicators (KPIs) are measurable values used to evaluate the success and efficiency of business operations. KPIs were developed using SQL calculations, DAX measures, and Power BI visualizations, focusing on five major business areas: Revenue, Passenger, Booking, Operational, and Route & Station Performance.

8.2 KPI Framework Overview

Business Area	KPI	Purpose
Revenue	Total Revenue	Measures financial performance
Revenue	Average Fare	Measures pricing level
Passenger	Total Passengers	Measures customer reach
Booking	Total Tickets	Measures booking volume
Booking	Cancellation Rate	Measures booking efficiency
Operations	Delay Percentage	Measures punctuality
Operations	Average Delay Time	Measures service reliability
Route	Top Performing Route	Identifies profitable routes
Station	Passenger Traffic	Measures station demand
Customer	Popular Travel Class	Understands passenger preference

8.3 Revenue Performance Analysis

KPI 1: Total Revenue

Total revenue represents the overall income generated from ticket sales.

```
Total Revenue = SUM(Bookings[total_amount])
```

Business Insight: Railway management should focus on expanding services on high-revenue routes, improving premium class facilities, and implementing demand-based pricing strategies.

8.8 Operational Performance Analysis

KPI 6: Delay Percentage

```
Delay Percentage =
DIVIDE(
    COUNT(Delays[delay_id]),
    COUNT(Train_Schedule[schedule_id])
)
```

Business Insight: Reducing delays improves passenger satisfaction, railway reputation, and operational efficiency.

8.12 Travel Class Analysis

Class	Passenger Volume	Revenue Contribution
Economy	High	Moderate
Premium	Lower	Higher

8.15 Major Business Insights

Insight: Revenue Concentration

Observation: A limited number of routes contribute a major portion of revenue.

Business Impact: Revenue optimization should focus on high-performing routes.

Insight: Seasonal Passenger Demand

Observation: Passenger bookings fluctuate throughout the year.

Business Impact: Seasonal planning can improve resource utilization.

Insight: Operational Delays Affect Service Quality

Observation: Repeated delays are concentrated in specific trains and routes.

Business Impact: Targeted improvements can increase punctuality.

Insight: Station Traffic is Uneven

Observation: Some stations handle significantly higher passenger volumes.

Business Impact: Infrastructure investment should prioritize busy locations.

Insight: Premium Classes Drive Revenue

Observation: Higher-class tickets generate more revenue despite lower volume.

Business Impact: Premium services represent an important revenue opportunity.

9. Business Recommendations and Project Conclusion

9.1 Introduction

Based on the findings from exploratory analysis, SQL analysis, and Power BI dashboard insights, several recommendations were developed to improve railway operational efficiency, revenue performance, and passenger satisfaction.

Recommendation 1: Optimize Train Scheduling Based on Passenger Demand

Observation: Booking trends show that demand varies significantly across routes and seasons.

Recommended Actions:

- Increase train frequency on high-demand routes.
- Introduce additional coaches during peak seasons.
- Reduce unnecessary services on consistently low-demand routes.

Expected Business Impact: Improved passenger satisfaction, better resource utilization, reduced overcrowding.

Recommendation 2: Improve Delay Management and Train Punctuality

Observation: Specific trains and routes experience frequent delays.

Recommended Actions:

- Monitor frequently delayed trains.
- Analyze root causes of delays.
- Improve preventive maintenance schedules.
- Optimize train scheduling.

Expected Business Impact: Improved punctuality, reduced passenger complaints, increased trust.

Recommendation 3: Focus on High-Revenue Routes

Observation: A small number of routes contribute significantly to total revenue.

Recommended Actions:

- Increase service frequency.
- Improve passenger facilities.
- Add premium services.
- Optimize ticket pricing.

Expected Business Impact: Increased revenue, better customer retention, improved route profitability.

Recommendation 4: Improve Station Infrastructure

Observation: High-traffic stations require focused resource allocation.

Recommended Actions:

- Increase staff availability.
- Improve waiting areas.
- Expand ticket counters.
- Improve passenger information systems.

Expected Business Impact: Reduced congestion, improved station experience, faster service delivery.

Recommendation 5: Implement Dynamic Pricing Strategies

Observation: Premium classes generate higher revenue despite fewer passengers.

Recommended Actions:

- Adjust prices based on demand.
- Offer discounts during low-demand periods.
- Optimize premium class pricing.

Expected Business Impact: Increased revenue, better seat utilization, improved profitability.

Recommendation 6: Enhance Digital Booking Experience

Observation: Digital payment methods represent a significant portion of transactions.

Recommended Actions:

- Improve mobile booking experience.
- Add personalized travel recommendations.
- Provide real-time notifications.
- Simplify payment processes.

Expected Business Impact: Higher customer satisfaction, increased digital adoption, reduced manual workload.

9.3 Operational Improvement Roadmap

Priority	Area	Recommended Action	Expected Benefit
High	Train Delays	Improve maintenance planning	Better punctuality
High	Revenue	Optimize high-performing routes	Increased income
Medium	Stations	Improve infrastructure	Better passenger experience
Medium	Scheduling	Demand-based planning	Resource optimization
Low	Analytics	Predictive modeling	Future decision support

9.4 Project Challenges

Data Quality Issues

Problem: Raw datasets contained missing values, duplicate records, and inconsistent formats.

Solution: Implemented missing value handling, duplicate removal, and data standardization using Python Pandas.

Dataset Integration

Problem: Multiple datasets had different structures requiring proper relationships.

Solution: Used primary/foreign keys, Pandas merge operations, and SQL JOIN operations.

Data Validation

Problem: Ensuring calculated metrics matched across Python, SQL, and Power BI.

Solution: Performed KPI validation and cross-checking aggregate values.

Dashboard Performance

Problem: Large datasets can slow Power BI performance.

Solution: Applied data modeling optimization, removal of unnecessary columns, and efficient DAX calculations.

9.5 Project Limitations

Historical Data Only

The analysis is based on historical records.

Impact: Real-time operational changes cannot be captured.

Limited External Factors

The dataset does not include weather conditions, traffic congestion, infrastructure issues, or external events.

Impact: Some delay causes cannot be fully explained.

No Predictive Analytics

The project focuses on descriptive analytics and does not include demand forecasting, delay prediction, or revenue prediction.

Impact: Forward-looking insights are limited.

9.6 Future Enhancements

Machine Learning Demand Forecasting

Predict future passenger demand using Linear Regression, Random Forest, and Time Series Forecasting.

Train Delay Prediction

Predict possible delays before they occur using historical delays, route information, weather conditions, and maintenance history.

Real-Time Analytics Dashboard

Connect live railway data sources using streaming data, APIs, and Azure/AWS services.

Predictive Maintenance System

Predict equipment failures using train sensor data, maintenance history, and operating conditions.

Customer Segmentation

Understand passenger groups using clustering algorithms and RFM Analysis.

9.8 Conclusion

This project demonstrates how data analytics can improve railway management by transforming operational data into meaningful business intelligence. Through Python-based preprocessing, SQL analysis, and Power BI visualization, the project provides insights into passenger behavior, revenue performance, booking patterns, route efficiency, station performance, and train delays.

The developed dashboard enables stakeholders to monitor important KPIs and identify opportunities for operational improvement. By applying data-driven decision-making, railway organizations can improve efficiency, optimize resources, increase revenue, and deliver better passenger experiences.

Project Outcome Summary

Area	Outcome
Data Processing	Cleaned and transformed raw railway datasets into analysis-ready data
Data Analysis	Identified trends in passengers, revenue, routes, and operations
SQL Analysis	Generated business insights using relational queries
Dashboard Development	Created interactive Power BI reports for decision-making
Business Insights	Provided recommendations for improving railway efficiency
Future Scope	Enabled opportunities for predictive analytics and automation

The Railway Management Analytics project showcases how data-driven approaches can transform complex railway operational data into actionable business intelligence, enabling better decisions, improved efficiency, and enhanced passenger services.

Project developed by Vyshnavi Katkam | Aspiring Data Analyst | Entry Level

10. Appendix

10.2 Project Folder Structure

```
Railway_Management_Analytics
|
+-- data
|   +-- raw
|   |   +-- passengers.csv
|   |   +-- bookings.csv
|   |   +-- tickets.csv
|   |   +-- trains.csv
|   |   +-- routes.csv
|   |   +-- stations.csv
|   |   +-- payments.csv
|   |   +-- delays.csv
|   |   +-- maintenance.csv
|   |   +-- employees.csv
|   |
|   +-- cleaned
|       +-- railway_management_system_final.csv
|
+-- python
|   +-- data_cleaning_analysis.ipynb
|
+-- sql
|   +-- railway_analysis_queries.sql
|
+-- powerbi
|   +-- Railway_Management_Dashboard.pbix
|
+-- documentation
|   +-- project_report.pdf
|   +-- data_dictionary.xlsx
|   +-- er_diagram.png
|
+-- images
+-- presentation
+-- README.md
+-- LICENSE
```

10.5 Power BI Documentation – Dashboard Pages

Dashboard Page	Purpose	Key Visuals
Page 1: Executive Overview	Management-level summary	Total Revenue, Tickets, Passengers KPIs; Revenue & Booking Trends
Page 2: Passenger Analysis	Customer behavior analysis	Age Distribution, Gender Analysis, City-wise Count
Page 3: Revenue Analysis	Financial performance	Monthly Revenue, Route Revenue, Class Revenue
Page 4: Operations Analysis	Operational efficiency	Delay Count, Average Delay, Train Performance
Page 5: Route & Station Analysis	Network performance	Top Routes, Station Traffic, Revenue Contribution

10.6 DAX Measure Documentation

Measure Name	DAX Formula
Total Revenue	SUM(Bookings[total_amount])
Total Tickets	COUNT(Tickets[ticket_id])
Total Passengers	DISTINCTCOUNT(Passengers[passenger_id])
Average Fare	AVERAGE(Tickets[fare])
Cancellation Rate	DIVIDE(COUNTROWS(FILTER(Bookings, Bookings[booking_status]="Cancelled")), COUNT(Bookings[booking_id]))

10.10 Resume Project Description

Railway Management Analytics Dashboard

Data Analyst Project | Vyshnavi Katkam

Developed an end-to-end railway analytics solution using Python, SQL, and Power BI. Cleaned and transformed 30,000+ operational records, performed exploratory analysis, created business KPIs, and developed interactive dashboards to analyze revenue, passenger trends, route performance, and train delays.

Technologies: Python | Pandas | SQL | MySQL | Power BI | Excel